

Preoperative electroacupuncture for postoperative nausea and vomiting in laparoscopic gynecological surgery: a randomized controlled trial

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Abstract

Objective: We aimed to evaluate the effectiveness and safety of preoperative electroacupuncture (EA) on the incidence of postoperative nausea and vomiting (PONV), and severity of postoperative pain, in gynecological patients undergoing laparoscopic surgery. The effects of EA administered at different preoperative time points were compared.

Methods: A total of 413 patients undergoing elective laparoscopic gynecological surgery were randomly allocated into 4 groups receiving EA the day before surgery (Group Pre, $n = 103$), 30 min before (Group 30, $n = 104$) or both (Group Comb, $n = 103$), or usual care alone (Group Usual, $n = 103$). All acupuncture groups had usual care. The incidence of PONV and pain at 24 h were primary outcomes. Secondary outcomes included the severity of postoperative nausea, vomiting and pain, requirement for antiemetic medication and quality of recovery (QoR)-15 scores after surgery.

Results: There were significant differences between the four groups in nausea and vomiting incidence (0–24 h), postoperative antiemetic use (0–48 h), and postoperative pain (0–6 h), with the EA groups recording the lowest levels. Regarding primary outcomes, incidence of nausea and vomiting at 6–24 h was 28/11/18/11% ($p = 0.003$) 23/5/8/9% ($p < 0.001$), respectively, for Groups Usual/Pre/30/Comb. Accordingly, EA reduced the incidence of nausea and vomiting at 6–24 h by 61/34/60% and 79/65/61% for Groups Pre/30/Comb, respectively. Regarding secondary outcomes, incidence of nausea and vomiting at 0–6 h was 20/9/11/7% ($p = 0.013$) and 17/7/9/6% ($p = 0.021$), respectively, for Groups Usual/Pre/30/Comb. Rescue antiemetics at 0–6 h were required by 18/4/11/4% ($p = 0.001$) in Groups Usual/Pre/30/Comb. The mean numerical rating scale (NRS) pain score (0–10) at 0–6 h was significantly different between groups (2.45/1.89/2.01/1.97 for Groups Usual/Pre/30/Comb, $p = 0.024$). There were no significant differences between the three EA-treated groups.

Conclusion: In gynecological patients undergoing laparoscopic surgery and treated with multimodal antiemetic methods, one session of preoperative EA may be a safe adjunctive treatment for PONV prophylaxis. Optimal timing of EA requires further verification.

Trial registration number: ChiCTR-INTR-16010035 (Chinese Clinical Trial Registry).

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Introduction

Postoperative nausea and vomiting (PONV) is a frequent and unpleasant complaint after anesthesia, affecting 20%–30% of patients,¹ and up to 80% of high-risk patients.² PONV and postoperative pain are undesirable for both patients and clinicians,³ and are the leading causes of unplanned admission following laparoscopic gynecological day surgery, leading to higher costs.⁴ Female sex of patients, laparoscopic surgery, gynecological procedures and opioid analgesia are all independent risk factors for PONV.^{5,6}

Drug therapies cannot fully relieve PONV and may cause side effects such as sedation and headache.^{7–9} Additional effective non-drug therapies for PONV and postoperative pain enable minimization of pharmacological antiemetics and analgesics, and associated side effects.

Traditional Chinese medicine holds that acupuncture supports homeostasis and minimizes the disturbances to normal function caused by surgery.^{10–12} Studies in human populations have shown that perioperative acupuncture minimized the elevation of serum serotonin,¹³ disrupted sympathoadrenal responses¹⁴ resulting from surgery, and reduced postoperative pain intensity and opioid analgesic consumption.^{15–17}

Acupuncture has been included in international management guidelines for PONV management as the only non-pharmacological intervention.^{9,18} A 2015 Cochrane systematic review of 59 trials found that stimulation at the traditional acupuncture point location PC6 (*Neiguan*) was significantly better than sham acupuncture and reduced both PONV incidence and the requirement for “rescue” antiemetic treatment by one third.^{19,20} Despite low-quality evidence, the review stated that further sham acupuncture trials would not change the conclusion that acupuncture effectively reduces PONV. It recommended that further trials should compare the effects of acupuncture plus multimodal antiemetic interventions with the latter alone.

Our pilot study in 40 patients at moderate to high risk of PONV found that, under best anesthesia practice, one in five patients at high risk still experienced PONV and that additional acupuncture could provide significant relief for this cohort.²¹ Acupuncture provided 24 h before surgery resulted in 25% and 75% reductions in postoperative nausea (PON) and postoperative vomiting (POV), respectively.

Few studies have evaluated the optimal timing or dosage of acupuncture on post-surgical recovery, and results have been inconsistent.^{12,22,23} Acupuncture was delivered 30 min prior to surgery in many trials,^{24–26} but may be stressful to patients as it is close to the time of surgery. Acupuncture one day before surgery produced better pain reduction than 30 min prior in a previous study²⁷ and was well accepted by both clinicians and patients in our feasibility study. It is not

known whether effectiveness would be increased when combined with a second session 30 min prior to surgery; however, two sessions were found to be necessary to alleviate chronic insomnia.²⁸

Herein, we conducted a single-center, four-arm randomized, controlled study in patients undergoing elective laparoscopic gynecological surgery with multimodal antiemetic prophylaxis. We aimed to: (1) evaluate the additional effects of preoperative electroacupuncture (EA) on PON and POV, and postoperative pain, compared with multimodal antiemetics alone; and (2) compare the effects of EA administered 24 h or 30 min before surgery, or a combination of both, on PONV and pain.

Methods

This study was reviewed and approved by the Human Research Ethics Committee of the Affiliated Hospital of Nanjing University of Traditional Chinese Medicine (2016NL-019-02), and registered with the Chinese Clinical Trial Registry at <http://www.chictr.org.cn/listbycreator.aspx> (ref. ChiCTR-INR-16010035) on 29 November 2016. All participants in the study were voluntary, understood the procedures, and provided written consent. The study hospital is a 2500-bed teaching hospital in Nanjing. More detailed methods are available in the published protocol.²⁹

Sample size calculation was based on the primary aim of this study, which was to evaluate the additional effects of preoperative EA on PONV, compared with modern multimodal antiemetics alone. According to our pilot study,²¹ we estimated that 75 participants per group would be needed to detect a 15% reduction in vomiting with 80% power using the following formula:

$$n = \frac{(1.96\sqrt{2pq} + 0.8416\sqrt{p_0q_0 + p_1q_1})^2}{(p_1 - p_0)^2}$$

$$\left(q_0 = 1 - p_0, q_1 = 1 - p_1, p = \frac{p_0 + p_1}{2}, q = 1 - p \right)$$

where n represents the sample size of usual care and EA groups, and p_0 and p_1 represent the incidence of vomiting in the two groups. Anticipating a 35% potential loss due to conversion to open surgery, we estimated that a total of 103 participants per group would be needed.

Participants and study design

Between December 2016 and February 2018, 420 patients scheduled for elective laparoscopic gynecological surgery

were screened for eligibility. The inclusion criteria were: American Society of Anesthesiologists (ASA) class I or II; general anesthesia; female sex; age 18–60 years; non-smoking; body mass index 20–30 kg/m²; undergoing laparoscopic hysterectomy, myomectomy, oophorectomy, salpingoopherectomy or ovarian cystectomy. Exclusion criteria were pregnancy or breastfeeding; psychiatric disorders; substance abuse; chronic pain syndromes; use of antiemetic, opioid or glucocorticoid medications within 3 days of surgery; nausea or vomiting within 24 h of surgery; cardiac pacemaker; infection near the planned needling sites; surgery lasting over 3 h; conversion to an open procedure; or intraoperative bowel damage. The number of PONV risk factors was calculated using the simplified Apfel score.³⁰ The four risk factors are: female gender; non-smoking status; history of PONV or motion sickness; and postoperative opioids (with a score of 0 representing no risk and 4 the highest risk). For every additional risk factor, the likelihood of PONV increases by 20%.

Eligible participants were randomly assigned to one of the following four groups using a computer-generated randomization sequence and secure code: (1) Group Pre, who received EA the day before surgery; (2) Group 30, who received EA 30 min before surgery; (3) Group Comb, who received EA both the day before and 30 min before surgery; and (4) Group Usual, who received usual care. EA was given on the ward the day prior to surgery, and in the Pre-anesthesia Care Unit 30 min before surgery, since surgical patients in China are routinely admitted to hospital 1–2 days prior to surgery. Each number was printed and stored in sequentially numbered, opaque, sealed envelopes. Eligible participants chose one envelope, which was opened by the acupuncturist who recorded the treatment allocation and provided the relevant procedure. Participants were instructed not to disclose their treatment.

Neither the anesthetists/surgeons nor the assessors were aware of the group allocation. Patients undergoing surgery on the same day were allocated to different rooms on the ward before surgery. All clinical outcome measures were conducted by a research nurse who was blinded to group allocation.

Acupuncture was administered at bilateral PC6, 2 *cun* or three finger breadths proximal to the wrist crease on the medial forearm, and ST36 (*Zusanli*), 3 *cun* or four finger breadths distal to the base of patella and one finger breadth lateral to the anterior border of the tibia.³¹ Point locations were selected based on Chinese medicine theory, previous trials and our feasibility study.^{21,32,33} After skin disinfection, stainless steel acupuncture needles (Suzhou Medical Supplies Factory Co. Ltd, 0.30x40 mm, China) were inserted by a doctor qualified in acupuncture and both Chinese and Western medicine, and manipulated until the patient reported *de qi* sensation (a radiating feeling of swelling or heaviness). The electrodes were connected from PC6 to ST36. An electric current was delivered for 30 min in dense-disperse mode (20–100 Hz) using a

stimulator (XS-998B, Nanjing Xiaosong Medical Instrument Research Institute, Nanjing, China) at a level where participants reported a strong but not painful sensation. Participants in the usual care group received no EA. All EA groups also received usual care.

Before induction of anesthesia and during surgery, electrocardiography, heart rate, pulse oximetry, non-invasive arterial pressure, bispectral index (BIS) and end tidal carbon dioxide (CO₂) were monitored continuously. General anesthesia was induced with intravenous midazolam 0.05 mg/kg, propofol 2–3 mg/kg, sufentanil 0.2–0.4 µg/kg, vecuronium 0.1 mg/kg, and lidocaine 1 mg/kg. Total intravenous anesthesia with continuous propofol and remifentanyl was used for anesthesia maintenance. Intermittent sufentanil and vecuronium were given as required. BIS ranged from 40 to 65. Volume-controlled ventilation was used and regulated to maintain end-tidal CO₂ between 4.3 and 5.1 kPa. Dexamethasone 5 mg and tropisetron 5 mg were administered at induction and immediately after skin closure, respectively, to prevent PONV in all groups. Intravenous parecoxib 40 mg was used for adjunctive pain management at the end of surgery. Patients were extubated when adequate spontaneous ventilation was established. Flurbiprofen 100 mg was administered when the numerical rating scale (NRS) pain score was >3, and metoclopramide 10 mg was infused as an antiemetic “rescue” therapy to any patient who experienced nausea, vomiting or both, and requested medication.

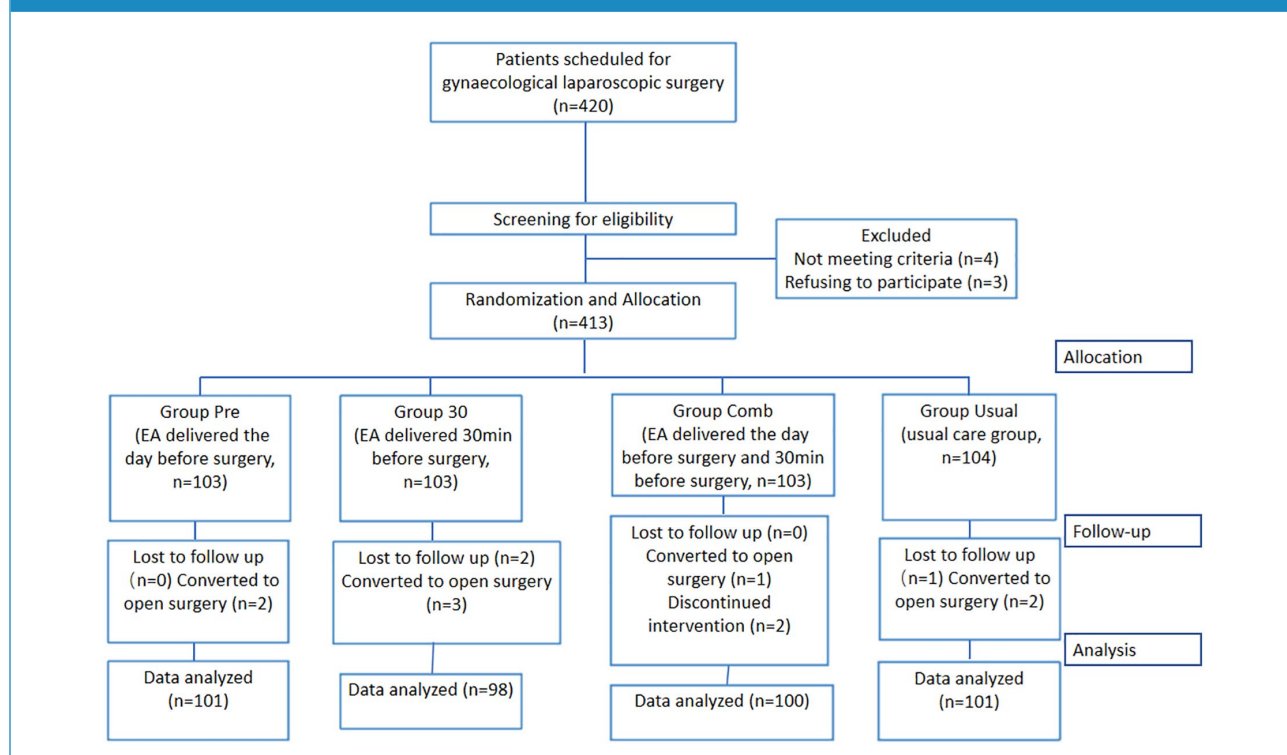
Outcome assessments

Patients were followed up while remaining in hospital 6, 24, 48 and 72 h after surgery. The primary outcome measures were the incidence of PON and POV at 6–24 h after surgery, and pain on movement at 24 h. All patients were asked “did you feel nauseous/did you vomit in the time period since the last consultation?” The rate of reduction in PON and POV was calculated as: (incidence in Group Usual – incidence in EA groups)/incidence in Group Usual. A NRS (0–10) was used to rate pain, with 0 indicating no pain, and 10 the worst possible pain.

The secondary outcomes included severity of PON and POV; number of patients requiring postoperative rescue antiemetics and analgesics at 0–6 h, 6–24 h, 24–48 h and 48–72 h after surgery; and incidence of PON and POV at 0–6 h, 24–48 h and 48–72 h. Severity of nausea and vomiting were separately assessed using a 4-point Likert-type severity scale, where 0=no nausea or vomiting, and 3=the worst imaginable nausea or vomiting, as in a previous study.³⁴

Other secondary outcome measures were the time interval between the end of surgery and first flatus, and the validated Quality of Recovery-15 (QoR-15) questionnaire.³⁵

Any adverse events of acupuncture such as pain, infection, bleeding and fainting during the procedure¹⁵ were recorded by the acupuncturist at the end of the treatment and on the following two days.

Figure 1. CONSORT (Consolidated Standards of Reporting Trials) flow diagram.

Missing data and statistical analysis

An expectation-maximization procedure was used to deal with missing data, as a method to “best guess the missing value based on the specific model and the existing data points.”³⁶

The Statistical Package for the Social Sciences (SPSS) for Windows version 19.0 (SPSS Inc., Chicago, IL, USA) was used to analyze the data. Normality of the data was tested by Kolmogorov–Smirnov analysis. Continuous variables with complete or approximately normal distribution are presented as mean $\pm$ standard deviation (SD). Chi-square test or Fisher’s exact tests were used to examine the differences between groups in categorical variables, and analysis of variance (ANOVA) or the Kruskal–Wallis test was used to compare descriptive continuous data among groups. A Bonferroni correction was performed when multiple paired comparisons were conducted. All reported *p* values were two-sided and $p < 0.05$ was considered statistically significant, except for the three primary outcome measures, for which the *p* value for statistical significance was adjusted to 0.017 to account for multiple testing.

Results

A total of 413 patients were enrolled and randomly allocated to the 4 groups, and 400 patients completed the study. Thirteen patients were excluded because they were lost to

follow-up ($n=3$), did not complete the acupuncture intervention ($n=2$), or converted to open surgery ($n=8$). The flow chart of the study is shown in Figure 1.

There were no differences between groups in preoperative status, Apfel scores, or surgery and anesthesia-related parameters (Table 1).

Primary outcomes: incidence of PON and POV, and pain at 24 h

The primary outcomes are shown in Table 2. Incidence of both nausea and vomiting peaked in all four groups between 6 h and 24 h postoperatively.

Between 6 and 24 h, Group Usual reported a nausea rate of 28% and vomiting rate of 23%, while nausea and vomiting rates in the EA groups Pre/30/Comb were 11/18/11% and 5/8/9%, respectively. Accordingly, EA reduced the incidence of nausea from 6 to 24 h by 61/34/60%, and the incidence of vomiting by 79/65/61%, for Groups Pre/30/Comb, respectively. The group differences were statistically significant (nausea: $p=0.003$, vomiting: $p < 0.001$).

Post hoc analyses showed that at 6–24 h the incidence of PON in Groups Pre and Comb were less than half that recorded in Group Usual (Pre vs Usual: $p=0.024$; Comb vs Usual: $p=0.024$, Bonferroni corrected). Incidence of POV in Group Pre and Group 30 was less than one third of that reported in Group Usual (Pre vs Usual: $p=0.002$, 30 vs Usual: $p=0.034$, Bonferroni corrected). There were no

Table 1. Baseline characteristics of patients.

Variables	Group Pre (n = 101)	Group 30 (n = 98)	Group Comb (n = 100)	Group Usual (n = 101)	p value
<i>Preoperative factors</i>					
Age (years; mean ± SD)	35.6 ± 8.1	34.7 ± 7.6	35.3 ± 8.2	35.7 ± 8.0	0.526
BMI (kg/m ² ; mean ± SD)	22.9 ± 3.13	22.7 ± 3.32	22.7 ± 2.96	22.97 ± 3.43	0.529
ASA physical status					
I	21	22	22	24	–
II	80	76	78	77	–
Apfel score (mean ± SD)	2.5 ± 0.50	2.4 ± 0.53	2.3 ± 0.48	2.28 ± 0.45	0.089
Surgical and anesthesia-related information					
<i>Type of surgery</i>					
Hysterectomy or myomectomy (n)	45	41	46	45	–
Salpingoophorectomy, oophorectomy or ovarian cystectomy (n)	56	57	54	56	–
Duration of surgery (min; mean ± SD)	63 ± 26.4	6.7 ± 26.4	6.9 ± 32.2	69 ± 32.8	0.275
Intraoperative sufentanil (μg; mean ± SD)	38.1 ± 4.97	39.1 ± 4.83	39.0 ± 5.12	37.8 ± 4.87	0.892
Intraoperative remifentanyl (mg; mean ± SD)	0.58 ± 0.23	0.58 ± 0.23	0.50 ± 0.25	0.57 ± 0.26	0.948
Intravenous fluid (mL; mean ± SD)	1090 ± 253.5	1096 ± 230.9	1104 ± 282.4	1121 ± 256.1	0.874

SD: standard deviation; BMI: body mass index; ASA: American Society of Anesthesiologists.

Table 2. Incidence and severity of postoperative nausea and vomiting, and incidence of requiring antiemetics after surgery.

Variables	Group Pre (n = 101)	Group 30 (n = 98)	Group Comb (n = 100)	Group Usual (n = 101)	F	DF	p value
Primary outcomes (6–24 h after surgery)							
Nausea (n)	11 [#]	18	11 [#]	27			0.003**
Vomiting (n)	5 ^{##}	8 [#]	9	22			<0.001**
Secondary outcomes							
0–6 h after surgery							
Nausea (n)	9	11	7 [#]	20			0.013*
Vomiting (n)	7	9	6	17			0.021*
Nausea (mean ± SD)	0.20 ± 0.68	0.22 ± 0.67	0.14 ± 0.53 [#]	0.46 ± 0.95	3.677	400	0.012*
Vomiting (mean ± SD)	0.16 ± 0.61	0.18 ± 0.63	0.11 ± 0.47 [#]	0.36 ± 0.79	2.851	400	0.037*
Antiemetic requirement (n)	4 [#]	11	4 [#]	18			0.001**
6–24 h after surgery							
Nausea (mean ± SD)	0.22 ± 0.66 [#]	0.34 ± 0.76	0.20 ± 0.60 [#]	0.50 ± 0.88	3.735	400	0.011
Vomiting (mean ± SD)	0.12 ± 0.53 [#]	0.15 ± 0.54 [#]	0.20 ± 0.67	0.41 ± 0.80	3.989	400	0.008
Antiemetic requirement (n)	8 ^{##}	16	9 ^{##}	28			<0.001**
24–48 h after surgery							
Nausea (n)	2	2	2	7			0.164
Vomiting (n)	1	2	1	5			0.307
Nausea (mean ± SD)	0.04 ± 0.28	0.03 ± 0.22	0.04 ± 0.28	0.12 ± 0.45	1.647	400	0.178
Vomiting (mean ± SD)	0.02 ± 0.20	0.02 ± 0.14	0.02 ± 0.20	0.08 ± 0.37	1.500	400	0.214
Antiemetic requirement (n)	1	2	1	7			0.045*

SD: standard deviation; PON: postoperative nausea; POV: postoperative vomiting.

*p < 0.05, **p < 0.01 between groups (*p < 0.025, **p < 0.005 for primary outcomes after correction); #p < 0.05, ##p < 0.01 when compared with Group Usual. At 48–72 h after surgery, the incidence of PON and POV was zero in all groups.

significant differences among the EA groups. Moreover, at 48–72 h after surgery, the incidence of PON and POV was zero in all groups (data not shown).

Severity of pain (Table 3) was mild over 72 h with highest levels at 0–6 h. Only a couple of patients in each group required analgesics. There was no difference between the three EA groups in intensity of pain on movement at 24 h.

Secondary outcome measures

Tables 2 and 3 list the results of secondary outcome measures. Incidence of nausea in Group Usual at 0–6 h was 21%, with 18% experiencing vomiting. EA groups Pre/30/Comb reported rates of nausea and vomiting of 9/11/7% and 7/9/6%, respectively. Accordingly, EA reduced the incidence of nausea and vomiting from 0 to 6 h by 57/46/66% and 61/48/66%, respectively, for Groups Pre/30/Comb when compared with Group Usual. The four groups were significantly different in incidence of nausea ($p=0.013$) and vomiting ($p=0.021$).

An intention-to-treat analysis was performed on PON and POV incidence at 0–6 h and 6–24 h including all 405 patients who were randomized and did not convert to open surgery (Supplemental Table 1). The results were similar to the per-protocol analysis as reported above (Supplemental Table 2).

Significant differences in the number of patients requiring rescue antiemetics were shown between the four groups at 0–6 h ($p=0.001$) and 6–24 h ($p<0.001$) with Group Usual having the highest rate.

Post hoc analyses showed that antiemetic requirements in Groups Pre and Comb were significantly lower than in Group Usual at 0–6 h (Pre vs Usual: $p=0.012$, Comb vs Usual: $p=0.013$), and 6–24 h (Pre vs Usual: $p=0.003$, Comb vs Usual: $p=0.007$).

There were statistically significant differences between the groups at 0–6 h ($p=0.024$) with those receiving usual care recording the highest pain levels. Scores in Group Pre were significantly lower than those in Group Usual ($p<0.05$; Table 3). No between-group differences were observed in pain at 48 or 72 h. There were no statistically significant differences between the three EA groups for any other secondary outcomes.

The time until passage of first flatus exhibited a significant inter-group difference ($p=0.003$) with Group Usual having the longest duration (Table 3).

No significant differences were seen between the four groups in severity of PON or POV, incidence of postoperative analgesic use, or QoR-15 scores.

Adverse events

Three patients in Group Comb experienced pain and itching at the needling sites, and two declined the second EA treatment. These minor and tolerable symptoms lasted <2 h, did not cause persistent visible signs, and did not

require medical attention. During the 3-day follow-up, no further adverse events were reported.

Discussion

In the present single-center, randomized, assessor-blinded clinical study among patients undergoing elective laparoscopic gynecological surgery, preoperative EA stimulation at ST36 and PC6 reduced the incidence of PON and POV at 6–24 h by 34%–61% and 61%–79%, respectively, and decreased rescue antiemetic use within 24 h. Early postoperative pain was reduced in the EA groups, as was the time to passage of first flatus. Postoperative pain was mild overall, therefore the differences in pain reduction between groups, while statistically significant for early pain, were not clinically relevant. The discussion hereafter therefore focuses on PONV.

No significant differences were detected between the three EA groups employing different timings of EA. However, when compared with Group Usual, Group Pre had a consistently lower incidence or severity of PON and POV and a lesser requirement for antiemetics at 6–24 h when PONV peaked.

Interpretation of findings: the combined intervention of EA and antiemetics

In our study, the incidence of PON and POV was 28% and 23%, respectively, at 6–24 h in the usual care group without EA, which is consistent with the rate of 25%–30% in high-risk patients receiving two intraoperative antiemetic drugs as prophylaxis reported in previous studies.³⁷ It is not ideal that more than one quarter of patients still experienced PONV despite a multimodal strategy.

Combined therapy with acupuncture and modern antiemetics may offer additional reduction for those patients. Indeed, the 2015 Cochrane review concluded that the effect of acupuncture was equivalent to antiemetics.²⁰ However, evidence is lacking as to whether a combination of acupuncture and antiemetics is better than standard antiemetics alone. Previous studies have provided conflicting evidence,^{20,38} and the Cochrane review called for comparative effectiveness studies to examine the additional effect of acupuncture. Our study aimed to address this specific concern.

The current study found that additional EA further reduced PONV and the requirement for rescue antiemetics. The relative reductions in Group Pre were 67%, 78% and 80% for PON, POV and medication use, respectively, compared to Group Usual. As the usual care group did not receive sham acupuncture, the reduction could be considered to be due to the placebo effect. The effect size seen in our study is arguably, however, too large to be explained simply by placebo. Very few studies have examined the difference between sham acupuncture and non-acupuncture.³⁹ One study⁴⁰ found, at 24 h, that PON, POV and requirements for rescue

Table 3. Severity of postoperative pain on movement (NRS), analgesic requirements, time to passage of first flatus after surgery and QoR score.

Variable	Group Pre (n = 101)	Group 30 (n = 98)	Group Comb (n = 100)	Group Usual (n = 101)	F	DF	p value
Primary outcomes (6–24 h after surgery)							
Pain at 6–24 h (mean $\pm$ SD)	1.15 $\pm$ 1.02	1.12 $\pm$ 1.03	1.17 $\pm$ 1.17	1.44 $\pm$ 1.30	1.642	400	0.179
Secondary outcomes							
Pain at 0–6 h (mean $\pm$ SD)	1.89 $\pm$ 1.47 [#]	2.01 $\pm$ 1.25	1.97 $\pm$ 1.25	2.45 $\pm$ 1.6	3.186	400	0.024*
Pain at 24–48 h (mean $\pm$ SD)	0.4 $\pm$ 0.76 [#]	0.54 $\pm$ 0.86	0.5 $\pm$ 0.96	0.73 $\pm$ 0.97	2.512	400	0.058
Pain at 48–72 h (mean $\pm$ SD)	0.19 $\pm$ 0.56	0.23 $\pm$ 0.59	0.19 $\pm$ 0.56	0.31 $\pm$ 0.70	0.847	400	0.469
Analgesics 0–48 h after surgery (n)	2	3	1	3			–
Time to passage of first flatus (h; mean $\pm$ SD)	18.38 $\pm$ 6.11 ^{##}	18.47 $\pm$ 6.47 [#]	18.82 $\pm$ 7.50 [#]	21.56 $\pm$ 8.07	4.615	400	0.003**
Overall QoR-15 (mean $\pm$ SD)	142.94 $\pm$ 6.82	144.74 $\pm$ 5.21	144.61 $\pm$ 5.35	144.23 $\pm$ 4.95	2.138	400	0.095

SD: standard deviation; QoR: quality of recovery; NRS: numeric rating scale.

*p < 0.05, **p < 0.01 between groups; #p < 0.05, ##p < 0.01 when compared with Group Usual.

medications were 35%, 10% and 25%, respectively, in the sham group and 30%, 25% and 20%, respectively, in the non-acupuncture group. The relative reductions following the sham intervention were –14%, 60% and –25%, respectively, which are much smaller than those seen in the current study.

Interpretation of findings: timing of EA

To our knowledge, no study to date has investigated the effects of two different timing schedules of preoperative EA, or compared one versus two sessions. Although we were not powered to examine equivalence, we found no suggestion that two sessions of EA were superior to one session of EA at any time point, and results were not significantly influenced by the timing of EA. Therefore, a single preoperative EA treatment is recommended. Based on our data, 350 patients would have been required in each group to provide definitive answers regarding the effect of EA timing. Optimal timing is also likely to be influenced by local factors including cost, availability of clinicians, time constraints and cultural acceptance. These considerations should be explored in future studies.

Gastrointestinal function is an important parameter for postoperative recovery and may promote the prevention of PONV, while PONV reduction may accelerate the recovery of intestinal ileus. EA has been shown to reduce the duration of postoperative ileus compared with sham after laparoscopic surgery for colorectal cancer.⁴¹ Therefore, EA may reduce PONV directly, or indirectly by regulating gastrointestinal function. Our study also supported the analgesic effects of preoperative EA, which was similar to our previous feasibility study and the conclusions of several systematic reviews.⁴²

The side effects attributable to EA in our study were minimal and resolved spontaneously. This is an important consideration, since the combination of different analgesics may lead to an increase in adverse effects. Therefore, preoperative EA may be a safe adjunctive antiemetic method.

Limitations

This study was limited to one hospital and patients undergoing a short duration surgery, and our findings may not extend to other procedures or longer durations. However, the adoption of standard anesthesia and EA procedures to patients undergoing a common surgery enhances the replicability of this study. In addition, since this study focused on patients who were at high risk of PONV, the results are clinically relevant.

We recorded the incidence of rescue analgesic drug use but not the dosage, which might potentially have shown differences between groups. The sample size was sufficient to observe the effectiveness of preoperative EA in PONV,

but not large enough to detect any differences between EA timing schedules.

Conclusion

Preoperative EA significantly decreases the incidence of PONV in patients undergoing laparoscopic gynecological surgery. One session of EA at ST36 and PC6 appears to be a safe adjunctive antiemetic method for PONV prophylaxis. The optimal timing of EA requires further investigation.

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Supplemental material

Supplemental material for this article is available online.

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